JavaScript Problem Solving

Set 5: Problems 41 to 50 (Strings, Arrays, Objects)

Is PDF ka istemal kaise karein

Har problem ko in 7 steps mein hal karo. Seedha code mat likho, pehle copy par logic banao:

1. Problem, Input, Output copy par apne alfaaz mein likho.
2. Ek chhota example khud kagaz par hal karo (insaan ki tarah).
3. Algorithm likho: numbered steps, Start se End tak.
4. Pseudocode likho: SET, FOR, IF, RETURN jaise words ke saath.
5. Dry run karo: table banao (i, current value, result).
6. Edge cases socho: khali input, ek element, negative numbers.
7. Ab JavaScript mein code likho aur test cases chalao.

Is set mein naya kya hai

- String ke methods: split, join, slice, repeat, toLowerCase (Problems 41-44).
- Array ke methods: map, filter, reduce, sort, flat (Problems 45-47).
- Objects ka array: item.price jaisi properties tak pahunchna (Problems 48, 49).
- Dono ka jor: strings ko words mein tod kar object mein ginna (Problem 50).
- Pehle sets mein built-in methods mana thay. Is set mein unhi ko seekhna maqsad hai.

Rules

- Pehle khud 20-30 minute koshish karo. Phir Hint 1, atko to Hint 2, phir Hint 3. Hints ek saath mat padhna.
- Is set mein built-in methods ki ijazat hai, bas har problem ke Constraints mein jo likha hai wahi follow karo. Jahan 'for loop se pehle karo' likha hai, wahan pehle loop se hal karo, phir method se.
- Function se hamesha return karo. console.log sirf test karne ke liye hai.
- Code ho jaye ya kahin atko to apna code Claude ko bhejo, wo hint ke saath check karega.
- Din mein 2-3 problems kaafi hain. Jaldi mat karo, har problem ka dry run zaroor karo. Set 4 pura karne ke baad ye set shuru karo.

Code bhejne se pehle checklist (aapki ab tak ki galtiyon se)

[ ] Loop: i < arr.length hai, <= to nahi? Agar arr[i + 1] use ho raha hai to i < arr.length - 1?

[ ] Comparison mein === hai, akela = to nahi?

[ ] return sahi jagah hai? Jawab sab elements dekhne ke baad chahiye to loop ke bahar, pehla match milte hi chahiye to loop ke andar.

[ ] Function return karta hai ya sirf print? Print karne se bahar wale console.log mein undefined aata hai.

[ ] Counter / total ki shuruaati value sahi hai? (sum ke liye 0, multiplication ke liye 1, sab se bada / chhota ke liye arr[0] ya -Infinity)

[ ] Khali input ("" ya []) par kya return hona chahiye? Wo PDF ki example table se dekho, apni marzi se nahi.

Quick Reference: Is set ke methods

Ye sirf yaad dihani ke liye hai. Har method ko pehle console.log mein try karo, phir problem mein lagao.

Method	Kya karta hai	Example
str.split(" ")	String ko array mein todta hai	"a b".split(" ") -> ["a", "b"]
arr.join("-")	Array ko string mein jodta hai	["a", "b"].join("-") -> "a-b"
str.slice(-4)	Aakhri 4 characters	"12345".slice(-4) -> "2345"
str.repeat(3)	String ko dohrata hai	"*".repeat(3) -> "***"
str.toLowerCase()	Sab letters chhote	"Hi".toLowerCase() -> "hi"
arr.reverse()	Array ulta (original badalta hai)	[1, 2, 3].reverse() -> [3, 2, 1]
arr.map(fn)	Har element par fn, naya array	[1, 2].map(x => x * 2) -> [2, 4]
arr.filter(fn)	Jin par fn true, wahi rakhta hai	[1, 2, 3].filter(x => x > 1) -> [2, 3]
arr.reduce(fn, start)	Sab ko ek value bana deta hai	[1, 2, 3].reduce((s, x) => s + x, 0) -> 6
arr.sort((a, b) => a - b)	Numbers ko ascending sort (original badalta hai)	[10, 9, 1].sort((a, b) => a - b) -> [1, 9, 10]
arr.slice()	Array ki copy	const copy = arr.slice()
arr.flat()	Ek level flatten	[[1], [2]].flat() -> [1, 2]
arr.includes(x)	x array mein hai ya nahi	[1, 2].includes(2) -> true

Dhyan: reverse() aur sort() original array ko badal dete hain. Original bachana ho to pehle arr.slice() se copy banao.

Problem 41: Reverse the Words

Level: Easy | Strings

New concept: split(), reverse() aur join(): string ko array bana kar wapas string banana.

DESCRIPTION

Given a sentence, return a new sentence with the order of the words reversed. Words are separated by a single space.

INPUT / OUTPUT

Input: A string, sentence.
Output: A string with the words in reverse order.

EXAMPLES

Input	Output	Note
"hello world"	"world hello"	
"I love JS"	"JS love I"	
"a b c"	"c b a"	
"one"	"one"	One word
""	""	Empty string

CONSTRAINTS

- Use split(), reverse() and join().
- Return a string, do not just print it.

STARTER CODE AND TEST CASES

```
function reverseWords(sentence) {
  // your code here
}

console.log(reverseWords("hello world"));  // "world hello"
console.log(reverseWords("I love JS"));    // "JS love I"
console.log(reverseWords("a b c"));        // "c b a"
console.log(reverseWords("one"));          // "one"
console.log(reverseWords(""));             // ""
```

Hints (sirf atko to dekhna, ek ek kar ke)

Hint 1: Sentence ko words ke array mein todne ke liye sentence.split(" ") likho. Pehle console.log karke dekho ke array kaisa banta hai.

Hint 2: Array ke methods ulta karne ke liye reverse(), aur wapas string banane ke liye join(" "). Teeno ko chain kar sakte ho: a.split(...).reverse().join(...).

Hint 3: Khali string par "".split(" ") ka jawab [""] hota hai (ek khali word wala array). Join karne par wapas "" hi milta hai, to special case ki zaroorat nahi. Kar ke dekho.

Problem 42: Longest Word

Level: Easy | Strings

New concept: split() se words nikalna, aur loop mein sab se lamba track karna.

DESCRIPTION

Given a sentence, return the longest word. If two words have the same length, return the first one. For an empty string return an empty string.

INPUT / OUTPUT

Input: A string, sentence.
Output: The longest word, as a string.

EXAMPLES

Input	Output	Note
"I love programming"	"programming"	
"cat dog"	"cat"	Tie: first one wins
"hi there you"	"there"	
"a"	"a"	One word
""	""	Empty string

CONSTRAINTS

- Use split() and a for loop.
- Do not use sort() or reduce().
- Return the word, do not just print it.

STARTER CODE AND TEST CASES

```
function longestWord(sentence) {
  // your code here
}

console.log(longestWord("I love programming"));  // "programming"
console.log(longestWord("cat dog"));             // "cat"
console.log(longestWord("hi there you"));        // "there"
console.log(longestWord("a"));                   // "a"
console.log(longestWord(""));                    // ""
```

Hints (sirf atko to dekhna, ek ek kar ke)

Hint 1: Pehle words ka array banao: sentence.split(" "). Phir ye Find Largest wala pattern hai, bas number ki jagah length compare karni hai.

Hint 2: Ek variable longest = "" rakho. Loop mein agar words[i].length > longest.length to longest = words[i].

Hint 3: Tie mein pehla word jeetne ke liye > use karo, >= nahi. Khali string par longest "" hi rehta hai, jo sahi jawab hai.

Problem 43: Make a URL Slug

Level: Easy | Strings

New concept: toLowerCase() aur split / join se spaces ko dash mein badalna.

DESCRIPTION

Given a title, return a URL slug: all letters in lowercase and the spaces replaced by dashes. Words are separated by a single space.

INPUT / OUTPUT

Input: A string, title.
Output: A slug string.

EXAMPLES

Input	Output	Note
"Hello World"	"hello-world"	
"My First Blog Post"	"my-first-blog-post"	
"A B C"	"a-b-c"	
"JavaScript"	"javascript"	One word
""	""	Empty string

CONSTRAINTS

- Use toLowerCase(), split() and join().
- Do not use a for loop for this one. Methods only.
- Return a string.

STARTER CODE AND TEST CASES

```
function slugify(title) {
  // your code here
}

console.log(slugify("Hello World"));          // "hello-world"
console.log(slugify("My First Blog Post"));   // "my-first-blog-post"
console.log(slugify("A B C"));                // "a-b-c"
console.log(slugify("JavaScript"));           // "javascript"
console.log(slugify(""));                     // ""
```

Hints (sirf atko to dekhna, ek ek kar ke)

Hint 1: Do kaam hain: sab letters chhote karna aur spaces ko "-" se badalna. Pehle ke liye toLowerCase() hai.

Hint 2: Spaces ko dash banane ka ek tareeqa: title.split(" ") se words ka array, phir us par join("-").

Hint 3: Bonus: replaceAll(" ", "-") se bhi ho sakta hai. Dono tareeqe try karo aur dekho ke kaunsa tumhein aasaan lagta hai.

Problem 44: Mask a Card Number

Level: Easy+ | Strings

New concept: slice(), repeat() aur string ki length se hisab lagana.

DESCRIPTION

Given a card number as a string, return it with every digit hidden by * except the last four. If the number has 4 or fewer characters, return it unchanged.

INPUT / OUTPUT

Input: A string of digits, num.
Output: A masked string of the same length.

EXAMPLES

Input	Output	Note
"1234567812345678"	"************5678"	12 stars, then last 4
"12345"	"*2345"	
"1234"	"1234"	4 digits: unchanged
"12"	"12"	Shorter than 4
""	""	Empty string

CONSTRAINTS

- Use slice() and repeat().
- Return a string, do not just print it.

STARTER CODE AND TEST CASES

```
function maskCard(num) {
  // your code here
}

console.log(maskCard("1234567812345678"));  // "************5678"
console.log(maskCard("12345"));             // "*2345"
console.log(maskCard("1234"));              // "1234"
console.log(maskCard("12"));                // "12"
console.log(maskCard(""));                  // ""
```

Hints (sirf atko to dekhna, ek ek kar ke)

Hint 1: Pehle chhota case: agar num.length <= 4 to seedha num return kar do.

Hint 2: Aakhri 4 characters nikalne ke liye num.slice(-4) (negative number ka matlab 'aakhir se ginti').

Hint 3: Stars ki tadaad num.length - 4 hai. "*".repeat(...) se utne stars banao aur aakhri 4 digits ke saath jodo (+).

Problem 45: Double the Evens

Level: Easy+ | Arrays

New concept: filter() aur map() ko chain karna.

DESCRIPTION

Given an array of numbers, return a new array that contains only the even numbers, each one doubled.

INPUT / OUTPUT

Input: An array of whole numbers, arr.
Output: A new array of doubled even numbers.

EXAMPLES

Input	Output	Note
[1, 2, 3, 4]	[4, 8]	2 and 4 doubled
[2]	[4]	
[-2, 0, 5]	[-4, 0]	Negatives and 0 work
[1, 3]	[]	No evens
[]	[]	Empty array

CONSTRAINTS

- Use filter() and map().
- Do not use a for loop.
- Do not change the original array.

STARTER CODE AND TEST CASES

```
function doubleEvens(arr) {
  // your code here
}

console.log(doubleEvens([1, 2, 3, 4]));  // [4, 8]
console.log(doubleEvens([2]));           // [4]
console.log(doubleEvens([-2, 0, 5]));    // [-4, 0]
console.log(doubleEvens([1, 3]));        // []
console.log(doubleEvens([]));            // []
```

Hints (sirf atko to dekhna, ek ek kar ke)

Hint 1: filter() un elements ko rakhta hai jin ke liye function true return kare. Even ke liye: x => x % 2 === 0.

Hint 2: map() har element par kaam karke naya array banata hai. Doubling ke liye: x => x * 2.

Hint 3: Dono ko chain karo: arr.filter(...).map(...). Pehle filter, phir map. Ye dono naya array dete hain, original ko nahi badalte.

Problem 46: Flatten a 2D Array

Level: Easy+ | Arrays

New concept: Array ke andar array: nested loop, aur phir flat() jaisa built-in method.

DESCRIPTION

Given an array whose elements are arrays, return one single array with all the inner elements in order (one level only).

INPUT / OUTPUT

Input: An array of arrays, arr.
Output: A new flat array.

EXAMPLES

Input	Output	Note
[[1, 2], [3], [4, 5]]	[1, 2, 3, 4, 5]	
[["a"], ["b", "c"]]	["a", "b", "c"]	Strings work too
[[], [1]]	[1]	An empty inner array
[[]]	[]	
[]	[]	Empty array

CONSTRAINTS

- First solve it with two nested for loops and push().
- Then solve it again with flat().
- Return a new array. Do not change the original.

STARTER CODE AND TEST CASES

```
function flatten(arr) {
  // your code here
}

console.log(flatten([[1, 2], [3], [4, 5]]));  // [1, 2, 3, 4, 5]
console.log(flatten([["a"], ["b", "c"]]));   // ["a", "b", "c"]
console.log(flatten([[], [1]]));             // [1]
console.log(flatten([[]]));                  // []
console.log(flatten([]));                    // []
```

Hints (sirf atko to dekhna, ek ek kar ke)

Hint 1: Bahar wala loop andar wale arrays par chalta hai (arr[i]), aur andar wala loop us ek array ke elements par (arr[i][j]).

Hint 2: Andar wale loop mein result.push(arr[i][j]) karo. Result ek khali array hai jo bahar banta hai.

Hint 3: Dusra tareeqa: arr.flat() ek line mein yehi kaam karta hai. Dono se kar ke dekho ke jawab ek jaisa aata hai.

Problem 47: Sort Numbers

Level: Medium-Easy | Arrays

New concept: sort() ka default behavior (alphabet ke hisab se) aur compare function. Original array ki hifazat.

DESCRIPTION

Given an array of numbers, return a new array sorted in ascending order. The original array must stay unchanged.

INPUT / OUTPUT

Input: An array of numbers, arr.
Output: A new sorted array.

EXAMPLES

Input	Output	Note
[10, 9, 1, 100]	[1, 9, 10, 100]	Not [1, 10, 100, 9]!
[3, -1, 2]	[-1, 2, 3]	Negatives
[2, 2, 1]	[1, 2, 2]	Duplicates
[5]	[5]	One element
[]	[]	Empty array

CONSTRAINTS

- Use sort() with a compare function.
- Do not change the original array.
- Return the new array.

STARTER CODE AND TEST CASES

```
function sortNumbers(arr) {
  // your code here
}

const nums = [10, 9, 1, 100];
console.log(sortNumbers(nums));            // [1, 9, 10, 100]
console.log(nums);                         // [10, 9, 1, 100] (unchanged!)
console.log(sortNumbers([3, -1, 2]));      // [-1, 2, 3]
console.log(sortNumbers([2, 2, 1]));       // [1, 2, 2]
console.log(sortNumbers([]));              // []
```

Hints (sirf atko to dekhna, ek ek kar ke)

Hint 1: Pehle try karo: console.log([10, 9, 1, 100].sort()). Jawab [1, 10, 100, 9] aata hai, kyunke default sort numbers ko text ki tarah (alphabet ke hisab se) sort karta hai.

Hint 2: Number ke hisab se sort ke liye compare function do: arr.sort((a, b) => a - b). Ye ascending order deta hai. Descending ke liye b - a.

Hint 3: sort() original array ko hi badal deta hai. Pehle us ki copy banao: arr.slice() (ya [...arr]) aur phir copy par sort() lagao.

Problem 48: Shopping Cart Total

Level: Medium-Easy | Objects

New concept: Array of objects: object ki properties (item.price) tak pahunchna.

DESCRIPTION

Given a shopping cart (an array of objects with name, price and qty), return the total cost: the sum of price multiplied by qty for every item.

INPUT / OUTPUT

Input: An array of objects { name, price, qty }, cart.
Output: One number: the total cost.

EXAMPLES

Input	Output	Note
[{ name: "pen", price: 10, qty: 3 }, { name: "book", price: 50, qty: 2 }]	130	30 + 100
[{ name: "a", price: 7, qty: 1 }]	7	One item
[{ name: "x", price: 5, qty: 0 }]	0	Quantity 0
[]	0	Empty cart

CONSTRAINTS

- Solve it with a for loop first.
- Then solve it again with reduce().
- Return the total, do not just print it.

STARTER CODE AND TEST CASES

```
function cartTotal(cart) {
  // your code here
}

const cart = [
  { name: "pen", price: 10, qty: 3 },
  { name: "book", price: 50, qty: 2 },
];
console.log(cartTotal(cart));                                    // 130
console.log(cartTotal([{ name: "a", price: 7, qty: 1 }]));       // 7
console.log(cartTotal([{ name: "x", price: 5, qty: 0 }]));       // 0
console.log(cartTotal([]));                                      // 0
```

Hints (sirf atko to dekhna, ek ek kar ke)

Hint 1: Ye array ke sum jaisa hi hai. Fark ye hai ke har element ek object hai, jis ki cheezein cart[i].price aur cart[i].qty se milti hain.

Hint 2: total = 0 se shuru karo. Har item ke liye total += cart[i].price * cart[i].qty.

Hint 3: reduce wala tareeqa: cart.reduce((sum, item) => sum + item.price * item.qty, 0). Aakhri 0 shuruaati value hai, jis se khali cart par 0 aata hai.

Problem 49: Names of Passed Students

Level: Medium-Easy | Objects

New concept: filter() se objects chunna, aur map() se unmein se ek property nikalna.

DESCRIPTION

Given an array of students (objects with name and marks), return an array with the names of the students who scored 50 or more.

INPUT / OUTPUT

Input: An array of objects { name, marks }, students.
Output: An array of names (strings).

EXAMPLES

Input	Output	Note
[{ name: "Ali", marks: 80 }, { name: "Sara", marks: 45 }, { name: "Ahmed", marks: 50 }]	["Ali", "Ahmed"]	50 passes
[{ name: "Bil", marks: 50 }]	["Bil"]	Exactly 50
[{ name: "Zed", marks: 10 }]	[]	Nobody passed
[]	[]	Empty array

CONSTRAINTS

- Use filter() and map().
- Return an array of names, not objects.

STARTER CODE AND TEST CASES

```
function passedNames(students) {
  // your code here
}

const students = [
  { name: "Ali", marks: 80 },
  { name: "Sara", marks: 45 },
  { name: "Ahmed", marks: 50 },
];
console.log(passedNames(students));                        // ["Ali", "Ahmed"]
console.log(passedNames([{ name: "Bil", marks: 50 }]));    // ["Bil"]
console.log(passedNames([{ name: "Zed", marks: 10 }]));    // []
console.log(passedNames([]));                              // []
```

Hints (sirf atko to dekhna, ek ek kar ke)

Hint 1: Pehle un students ko chuno jin ke marks 50 ya zyada hain: students.filter(s => s.marks >= 50).

Hint 2: Abhi tumhare paas pass hone walon ke poore objects hain. Sirf naam chahiye, to map() lagao: s => s.name.

Hint 3: Chain karo: students.filter(...).map(...). Dhyan rakho ke shart >= 50 hai, > 50 nahi.

Problem 50: Word Count Object

Level: Medium | Strings + Objects

New concept: Problem 35 ka counter pattern, ab words par. Aur khali string ka chhupa hua trap.

DESCRIPTION

Given a sentence, return an object that shows how many times each word appears. The words are the keys. The check is case-sensitive. Words are separated by a single space. For an empty string return an empty object.

INPUT / OUTPUT

Input: A string, sentence.
Output: An object like { the: 2, cat: 1 }.

EXAMPLES

Input	Output	Note
"the cat the dog"	{ the: 2, cat: 1, dog: 1 }	
"go go go"	{ go: 3 }	
"Go go"	{ Go: 1, go: 1 }	Case matters
"a"	{ a: 1 }	One word
""	{}	Empty string: be careful

CONSTRAINTS

- Use split() and a for loop.
- Do not use reduce().
- Return the object, do not just print it.

STARTER CODE AND TEST CASES

```
function wordCount(sentence) {
  // your code here
}

console.log(wordCount("the cat the dog"));  // { the: 2, cat: 1, dog: 1 }
console.log(wordCount("go go go"));         // { go: 3 }
console.log(wordCount("Go go"));            // { Go: 1, go: 1 }
console.log(wordCount("a"));                // { a: 1 }
console.log(wordCount(""));                 // {}
```

Hints (sirf atko to dekhna, ek ek kar ke)

Hint 1: Pehle words ka array banao (split). Phir Character Frequency wala pattern lagao, bas character ki jagah word.

Hint 2: Ek khali object result = {} banao. Har word ke liye: agar result[word] === undefined to result[word] = 1, warna result[word]++.

Hint 3: Trap: "".split(" ") ka jawab [""] hota hai (ek khali word), is liye result mein "": 1 aa jayega. Function ke shuru mein khali string ka case pehle sambhal lo.

Progress Tracker

Har problem ke saamne jo step mukammal ho, wahan tick (X) lagao.

#	Problem	Algorithm	Pseudocode	Dry run	Code	All tests pass
41	Reverse the Words					
42	Longest Word					
43	Make a URL Slug					
44	Mask a Card Number					
45	Double the Evens					
46	Flatten a 2D Array					
47	Sort Numbers					
48	Shopping Cart Total					
49	Names of Passed Students					
50	Word Count Object					

Bonus sawal (har problem ke baad khud se poocho)

- Is code ki time complexity kya hai? Kitni baar loop chalta hai? (Nested loop ho to kitni baar?)
- Is code ki space complexity kya hai? Kya naya array ya string bana?
- Best case aur worst case mein kya farq hai?
- Kya main isi problem ko doosre tareeqe se bhi hal kar sakta hoon?